Kai Sims

Carlsbad, 92009 | 760-707-9603 | kaijpsims@gmail.com | linkedin.com/in/kai-j-sims

PROFESSIONAL EXPERIENCE

UsCalibration, Irvine, CA June 2026 - Present

Engineer

- Calibrate aerospace, manufacturing, and medical equipment while maintaining measurement accuracy, traceability, and ISO-aligned quality documentation
- Write and standardize technician-facing calibration procedures, translating equipment-specific requirements into repeatable workflows that meet ISO quality standards
- Develop software-based tools and internal tracking systems for equipment location, storage status, field deployment, documentation, and technician workflow organization

HNTB, Oakland, CA May 2025 – August 2025

Intern

- Performed airflow analysis, pressure drop calculations, and pipe load assessments for HVAC, plumbing, and fire protection systems across four transit stations using Revit, AutoCAD, and Pipe-Flo
- Modeled worst-case operating scenarios to evaluate system capacity, mechanical loading, and code compliance
- Generated technical documentation and reports detailing system performance, safety margins, and integration constraints for multidisciplinary review

Structural Integrity Engineering, Chatsworth, CA June 2024 – September 2024

Intern

- Conducted 3D linear static and modal FEA in FEMAP on aircraft structural assemblies under maneuver, landing, and operational load cases
- Evaluated stress concentrations, displacement fields, and factor-of-safety margins; developed MATLAB scripts to automate test-data reduction and analytical correlation

PROJECT EXPERIENCE

Mechanical System Design, Escondido, CA June 2023 – Present

Intern

- Designed six custom suspension components in SolidWorks, analyzing bending, torsional, and shear loading to optimize stiffness and durability
- Fabricated components using CNC machining, sheet metal forming, MIG welding, and additive manufacturing, translating CAD models into testable hardware
- Diagnosed structural and drivetrain failures in performance vehicles through systematic load-path evaluation, physical inspection, and hands-on repair

Autonomous Nano-Scale 3D Printer Manufacturing Line, Berkeley, CA August 2025 – May 2026

Design Leader

- Designed precision mechanical fixtures, fluid handling assemblies, and motion subsystems for automated ink cartridge filling with micron-level alignment requirements
- Developed Python-based automation to control stepper-driven motion systems, integrate sensors, execute sequencing logic, and log manufacturing data
- Validated process stability through iterative testing, root cause isolation, tolerance control, and controlled parameter variation to improve repeatability and yield

Endoscopic Clip Applier Design, La Jolla, CA November 2024 – March 2025

Design Leader

- Owned mechanical architecture and tolerance stack-ups for a ligation device compatible with the da Vinci robotic surgical platform
- Built and tested twelve prototypes across eight design iterations to refine actuation force, friction interfaces, and alignment reliability
- Conducted structured mechanical testing to validate repeatability, durability, and ergonomic performance

EDUCATION

University of California, Berkeley May 2026
Master of Engineering in Mechanical Engineering 3.9 GPA

University of California, San Diego Mar 2025
Bachelor of Science in Mechanical Engineering 3.7 GPA

SKILLS

- **Design & Modeling:** SolidWorks, AutoCAD, Revit, KeyCreator
- **Analysis & Simulation:** FEMAP, MATLAB, Pipe-Flo
- **Fabrication & Prototyping:** CNC machining, MIG welding, 3D printing, metal/wood lathe operation
- **Programming & AI Tools:** Python, data analysis, AI-assisted design and automation